

HW 05

Lecture 5

Due: Tuesday, June 30 at 11:59 PM

AI use

Refer to the syllabus for our AI use policy. You will be asked to include any AI prompts used in this assignment at the end.

Starting a new Quarto file

For this assignment you will begin a new Quarto file. To do this, go to RStudio and click File > New File > Quarto Document. A pop-up will appear asking for a title, author, and format. You may leave these as the default options for now. Click OK and a new file will open. Update the YAML with your title, name, and date.

Exercise 1

Air pollution may cause lung deficiencies and development of diseases such as emphysema or COPD. A researcher is interested in the effect of air pollution on mice, and conducts an experiment whereby the researcher exposes mice to high levels of air exhaust for six hours a day for three weeks, and at the end of the three week period, evaluate whether the mice have developed deficiencies in lung function. In a sample of the mice used, 25 out of 100 developed such deficiencies. The researcher is interested in the number of mice that must be sampled before he obtains ten abnormal mice for use in further studies and proposes using a binomial distribution to model this quantity. Evaluate whether the assumptions for a binomial distribution are satisfied.

Exercise 2

Vaccination is one of the most effective public health tools ever created to control the spread of infectious, communicable diseases, but unfortunately recently there has been some pushback from the public against mandatory childhood vaccinations. Due to vaccine hesitancy, formerly well-controlled diseases such as measles have made a comeback in certain communities due to lack of vaccination and herd immunity. A researcher is interested in the number of children who develop measles across various school districts in the United States and proposes using a Poisson distribution to model this quantity. Evaluate whether the assumptions for a Poisson distribution are satisfied.

Exercise 3

A team of scientific researchers in Antarctica is investigating the number of eggs laid in a given season to Adelie penguin breeding pairs. In their initial survey of 100 randomly chosen breeding pairs, they found a mean of 2 eggs (standard deviation 0.3 eggs). Explain

why it would or would not be appropriate to use a Poisson distribution to model this quantity.

AI Attestation

Was AI used on this assignment? If so, describe your prompts below. If not, simply state “AI was not used on this assignment.”

Submission

As you’ve seen previously, we can **Render** into an .html file that can be opened by any web browser. To export it as a .pdf, open the file in your web browser and then print to or save as a .pdf document. Contact me in Ed Discussion if you need help!

You will submit the PDF documents for labs and homework to Gradescope as part of your final submission.

To submit your assignment:

- Access Gradescope through the menu on the BIOS 600 Canvas site.
- Click on the assignment, and you’ll be prompted to submit it.
- Mark the pages associated with each exercise. All of the pages of your lab should be associated with at least one question (i.e., should be “checked”).
- Select the first page of your .PDF submission to be associated with the “Formatting” section.

Grading

Component	Points
Ex 1	3
Ex 2	3
Ex 3	2
AI Attestation	1
Formatting	3

The “Formatting” grade is to assess the document format. This includes having a neatly organized document (no excessive output, warnings/messages when loading packages and/or data) with readable code and your name and the date updated in the YAML.